

Understanding Solar-Thermal Systems

by Justin Mizany

Why should homeowners invest in a solar-thermal system? It's all about economics.

With rising energy prices, many homeowners are now thinking of adding a solar component to their home, but they may feel priced out by the cost of a PV system. For them, a solar-thermal system—used to heat water—is a good economic option. Solar thermal is now even more affordable, thanks to a new 30% federal tax credit, which will be offered on all solar systems installed before December 31, 2007. For residential applications, there is a \$2,000 cap; there is no cap for commercial applications. The complete cost of the system (equipment and installation) may be applied to the credit, although the credit applies to the basis remaining after any state or utility incentive available to the taxpayer is taken. For solar water-heating systems, any property qualifying for the credit must be Solar Rating and Certification Corporation (SRCC) certified, or be certified by another comparable agency endorsed by the state. In addition, at least 50% of the energy used by the system to heat water must come from the sun. The average installed cost of a residential solar water heater in California is approximately \$7,000; therefore the \$2,000 cap on the federal solar tax credit covers roughly 30% of the average installed cost.

SOLAR SAVINGS

These incentives are only the beginning of the economic savings



SOLAR DEPOT

This residence in Vallejo, California, features a grid-tied solar electric system and a solar thermal system.

that a homeowner will typically realize with a solar-thermal system. According to the California Public Utility Commission, electricity rates have been increasing by an average of 6% per year over the past 35 years, and the wholesale price of natural gas has more than tripled in the past eight years. Whether utility customers have electric or gas water heaters, their operating costs will continue to rise over time—unless they use solar energy for water heating. One study by the California Energy Commission (CEC) shows the economic benefits of solar, taking into account the annualized costs of purchasing, installing, maintaining, and operating different types of water heater over a 30-year period (see Figure 1). This economic analysis includes the cost of financing as well as interest deductions, and is included in a conventional mortgage for new construction.

At current prices, the typical payback for a solar water-heating system in Northern California is eight to ten years, while a solar pool-heating system has a typical payback of

two to four years. The most impressive savings come in the form of avoided energy costs in the future, after the system has paid for itself.

SOLAR-THERMAL CHOICES

To take best advantage of a solar-energy system, a good southern exposure, with minimal shading, is essential. Rooftop installations are the most common, but the collectors can also be mounted on accessory buildings, or on ground mounts. In California, most solar water-heating systems can easily meet 100% of a household's water-heating needs in the summer (except, perhaps, along the foggy coast), and 20%–30% in the winter. On an average annual basis, solar water-heating systems are typically designed to meet 60%–70% of a household's annual hot water needs. Because we need hot water in the winter as well as the summer, solar water-heating systems use a gas or electric backup on cloudy days. In Northern California, gas backups are most common.

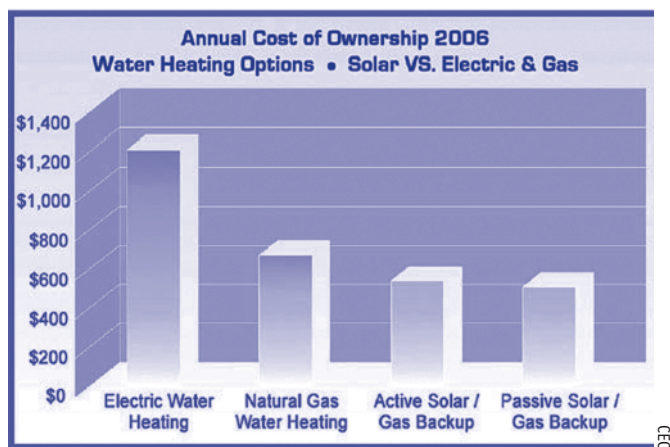


Figure 1. At current prices, the typical payback for a solar water-heating system in Northern California is eight to ten years.

Table 1. Pool Heating

Type of Pool Heater	Gas	Solar
Initial Cost	\$2,600	\$4,500
5-Year Operating Cost	\$6,000	\$ 0
Total Cost Over 5 Years	\$8,600	\$4,500

Solar water-heating systems come in two basic designs: active and passive. Active-solar systems have circulating pumps and controls and typically use flat-plate solar panels. These solar panels employ insulated metal boxes, copper pipes with black absorber plates, and glazing to produce water temperatures of 120°F–140°F for domestic use. Because active systems circulate water through the collectors, they are better suited for producing large volumes of hot water than are the passive-type systems.

Passive-solar systems have no pumps, controls, or moving parts. Passive systems come in two basic designs. One is a batch type integrated collector storage (ICS), which operates on inline water pressure to move the hot water from the ICS to the domestic storage tank. The other is the thermo siphon system, which has a separate panel and a well-insulated storage tank. Passive-solar systems use the natural flow of gravity to move the hot water into a solar storage tank located above the panel on the roof. They typically produce 60–80 gallons of hot water per day. Because these systems are so simple, they are often the most reliable and cost-effective means of providing

solar water heating on a small scale. Multiple units can be connected together to make more hot water.

STILL MORE SYSTEM OPTIONS

A solar pool-heating system can be very cost-effective, but overall savings will depend on the site. The typical payback period is two to four years. Several factors affect the savings realized with a solar pool-heating

system. These include the climate, the size of the pool, the cost of natural gas, the pool temperature to be maintained, the length of the swimming season, and whether or not there is a pool cover. A utility company study monitoring the amount of energy needed to heat a 500 ft² pool in Santa Clara, California, from May through September at a constant temperature of 80°F found that this system used approximately 1,200 therms of natural gas. At residential summertime tiered gas rates, this would come to about \$1,200 per year. For the same pool, a 400 ft² solar pool-heating system would maintain the same or higher temperatures through the same swimming season—with no cost for heating the water (see Table 1). This system easily achieved a four-year payback.

It is also possible to heat the interior of a home with a solar water-heating system. The most efficient way to accomplish this is with an in-floor radiant-heating system. These systems circulate hot water through pipes under the floor to warm the floor and heat the space. This is one of the most efficient ways to heat a home. It's clean and quiet, with no fans or ducts to stir up dust or pathogens,

allowing the homeowner to maintain a healthy indoor environment. While in-floor radiant-heating systems are more expensive to install—2–3 times the installation cost of forced-air heating—the annual operating cost is approximately 30%–40% less than that of a conventional forced-air heating system. The marginal increase in the installed cost of radiant heating compared to the installed cost of conventional forced-air heating has a five to six year payback. It is possible to save even more with solar-assisted in-floor radiant heating.

SMART SOLAR HOME ECONOMICS

The CEC estimates that the average household spends 30% or more of its energy bill on heating water, and that heating water accounts for more than 25% of the total energy used in a typical single-family home. Homeowners who are currently heating water using electricity can save 60%–70% on the cost by installing a solar water-heating system, according to CEC estimates (see Table 1). When you weigh these costs against those of a properly designed and installed solar water-heating system—which usually performs well for 20–40 years—the choice becomes quite simple.

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For more information:

To learn more about solar radiant heating, go to www.solardepot.com/pdf/2006/RadiantFloorOperatingCost.pdf.

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